EX.No.-2.2: CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

Date:14.10.2026

AIM:

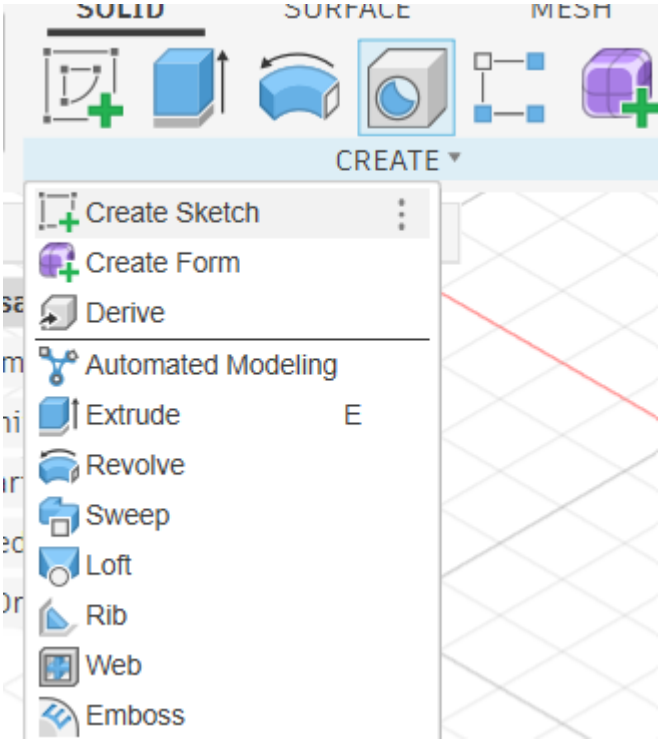
To construct a 3D solid model from the given isometric drawing using Autodesk Fusion 360.

REQUIREMENTS:

Autodesk Fusion 360
Windows 10 operating system
16 GB RAM (integrated graphics recommended 1 GB or more)
Internet connection with 2.5 Mbps download and 500 Kbps upload speed

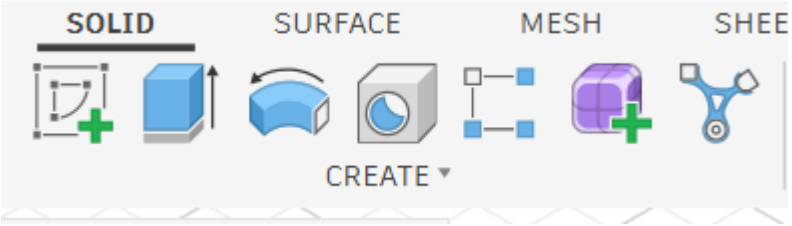
PROCEDURE:

1: Open Fusion 360
Launch Autodesk Fusion 360 and create a new design. Save the file with a suitable name.
2: Choose the Sketch Plane
Click on Create Sketch and select the XY plane to begin drawing the base profile.



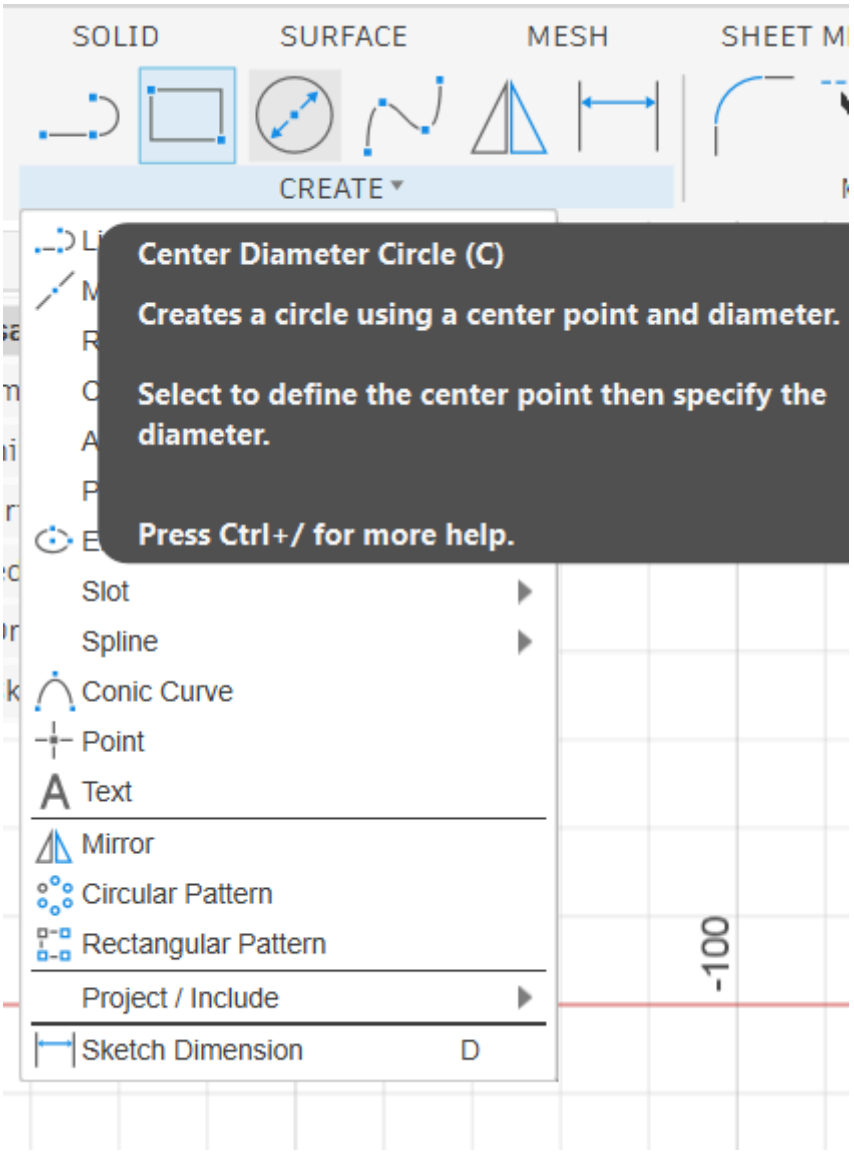
3: Draw the Base Profile
Choose Rectangle → Center Rectangle and place the rectangle at the origin. Set the width to 56 mm and height to 18 mm. Use the Center Point Arc tool to create the required arc with a radius of 18 mm. Then select Center Point Circle and draw the circle with a diameter of 16 mm. Click Finish Sketch.

4: Extrude the Base
Select the completed sketch and use the Extrude command. Enter 8 mm as the extrusion distance and press Enter to create the base solid.
5: Create a New Sketch
Select Create Sketch and choose the top face of the base model as the sketch plane.
6: Draw the Rectangular Profile
Select Rectangle → Center Rectangle and draw a rectangle with a width of 40 mm and a height of 10 mm. Complete the sketch using Finish Sketch.
7: Extrude the Rectangle
Select the rectangular profile and apply the Extrude command. Enter 32 mm as the extrusion distance and press Enter.



8: Construct the Triangular Profile
Create a new sketch on the required face. Select the polygon or triangle tool and draw the triangular profile with the specified dimensions of 32 mm and 12 mm. Click Finish Sketch.
9: Extrude the Triangle
Select the triangular profile and use Extrude. Set the extrusion distance to 6 mm in both directions and press Enter to complete the feature.
10: Draw the Polygon Profile
Start another sketch on the required face. Select Sketch → Polygon → Circumscribed Polygon. Set the polygon size according to the given dimensions of 40 mm and 30 mm. Choose 3 sides as specified and finish the sketch.
11: Extrude the Polygon
Select the polygon profile and apply the Extrude command. Enter 10 mm as the extrusion distance and press Enter.
12: Create the Circular Profile

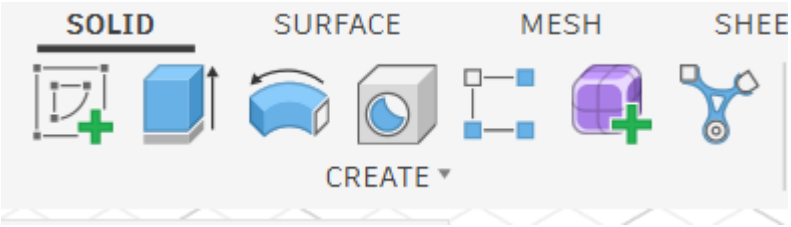
Start a new sketch on the required end face of the model. Choose Center Point Circle and draw the circular profile using the specified inner and outer diameters of 20 mm and 30 mm.



Click Finish Sketch.

13: Extrude the Circular Profile

Select the circular profile and apply the Extrude command. Set the extrusion distance to 10 mm in both directions and press Enter to create the circular feature.

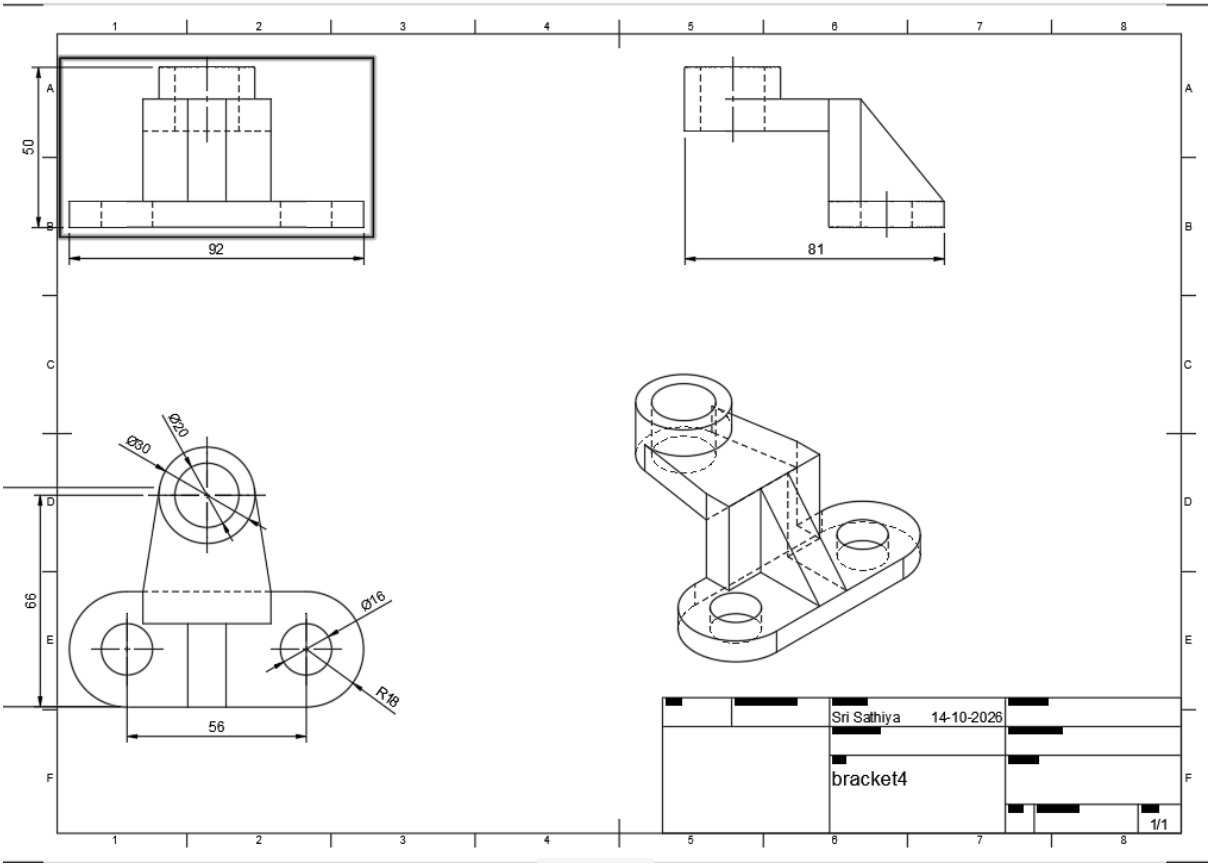


14: Complete the 3D Model

Check all the created features and dimensions. If necessary, use the Modify tools to make corrections. Adjust the view to an isometric position and save the completed design.

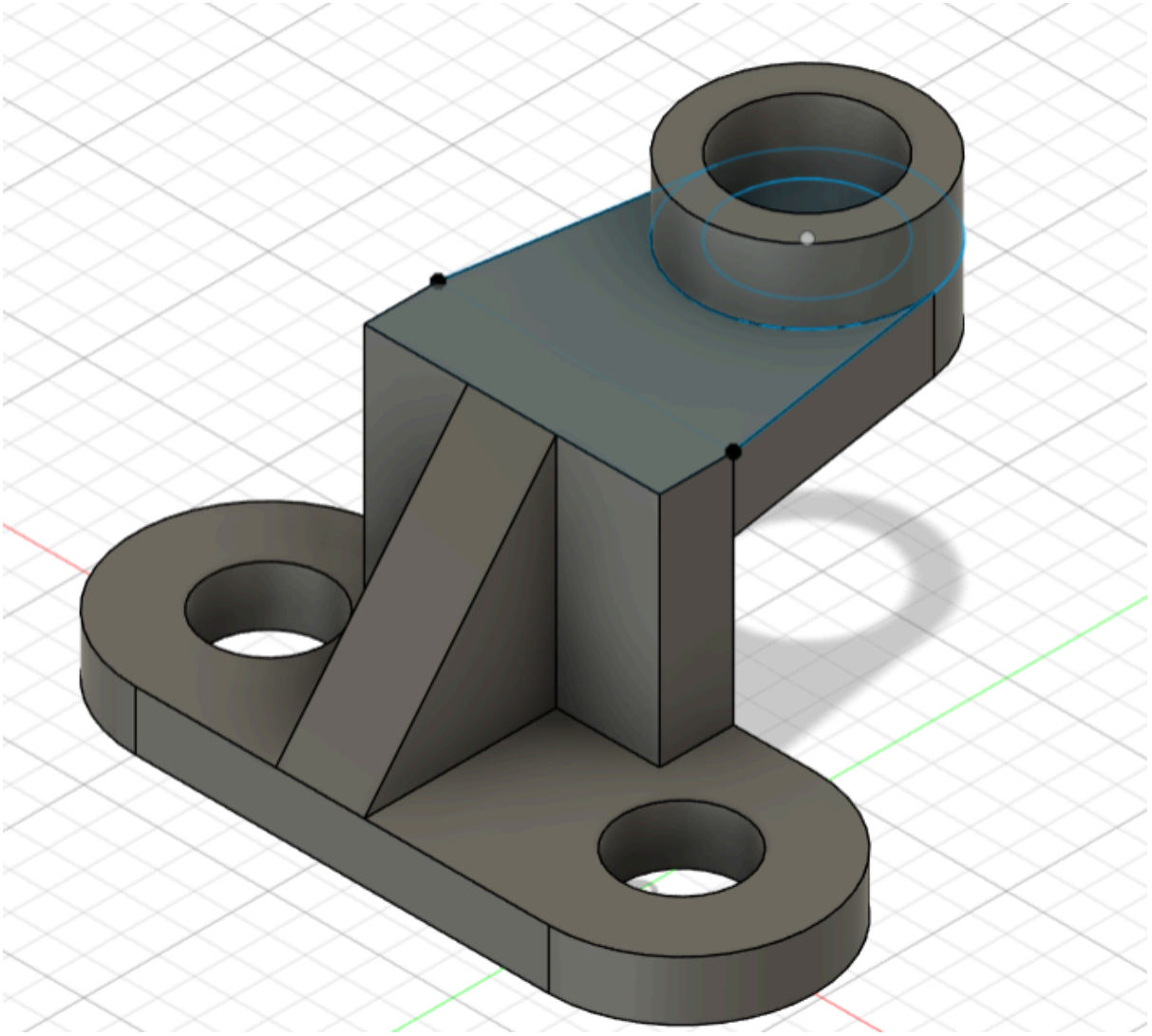
INPUT:

Given isometric drawing.



OUTPUT:

3D solid model created using Autodesk Fusion 360.



RESULT:

Thus, the 3D solid model was created from the given isometric drawing using Autodesk Fusion 360.